

Annie Prasanna Manoharan

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EDUCATION

University of Arkansas

Aug 2024 – May 2026

Master of Science in Computer Engineering

Fayetteville, AR

Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College

Aug 2019 – May 2023

Bachelor of Engineering in Electronics and Communication

India

TECHNICAL SKILLS

Machine Learning and AI: PyTorch, Computer vision, Scikit-learn, Flask, Pandas, NumPy

Cloud and Tools: Object detection and tracking, Git, Test automation, TensorFlow, CI/CD, Docker, Linux, Data validation and quality, Grafana, Jupyter, LSTM, Hyperparameter tuning, Cross-validation, GitHub Actions

Languages: Python, SQL

Data Engineering: ETL/ELT

PROJECTS

MiGa - Multi-Chicken Gait Assessment (AICV Lab)

YOLO11n-Pose, OC-SORT, LSTM, PyTorch

- Integrated YOLO11n-Pose for keypoint estimation, applied OC-SORT for multi-object tracking, and constructed a many-to-one LSTM sequence network for classification
- Built a data engineering framework to preprocess and segment raw frames into a detection split of 4,569 frames with 11,422 bounding boxes and a regression split of 55 expert-annotated sequences
- Implemented spatial normalization into local coordinate spaces to eliminate translational variance, achieving keypoint mAP 91.83% at 1,250 FPS, 0.93 MOTA, and 0.91 IDF1

Resume Classification Model

Python, Scikit-learn, Pandas, TF-IDF, Logistic Regression, Flask

- Built an end-to-end NLP classification pipeline using TF-IDF feature extraction and Logistic Regression across 25 job-role categories from 962 unstructured resumes
- Achieved 89.48% test accuracy and 88% macro F1 across 193 held-out test samples
- Deployed the model through a Flask REST API for real-time prediction

PROFESSIONAL EXPERIENCE

Data Engineering Intern

Jul 2024 – Present

InsOps Inc.

Chicago, Illinois

- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics
- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics

Teaching Assistant

Sep 2024 – May 2026

University of Arkansas, Dept. of Computer Science and Engineering

Fayetteville, AR

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals

Research Assistant

Sep 2024 – Jan 2026

AICV Lab, University of Arkansas

Fayetteville, AR

- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID

Project Engineer

Jun 2023 – Jun 2024

Soliton Technologies

India

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation
- Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements